

SPRINGFIELD NUCLEAR POWER PLANT

Primary Coolant System - Operations Overview

Reactor Operations

1. Normal operating range

The primary coolant loop operates between 270 and 290 degrees Celsius at full load. Temperatures below 250 degrees indicate a startup or low-load condition and are not by themselves a fault.

2. Alarms

The control room alarm sounds at 300 degrees Celsius. The alarm is advisory: it exists to give the shift crew time to identify the cause before a shutdown threshold is reached, and does not itself require a scram.

3. Instrumentation

Loop temperature is measured redundantly at the hot leg and the cold leg. A disagreement greater than 8 degrees between the two channels is treated as an instrument fault and reported to maintenance.